

# BlinkIT Grocery Sales Analysis

## Data Analytics Portfolio Project

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Tools: SQL (MySQL) | Power BI

Year: 2026

### 1. Project Overview

This portfolio project analyzes BlinkIT grocery sales data to evaluate overall business performance and identify key factors influencing sales, customer satisfaction, product performance, and outlet performance.

Using SQL (MySQL) for data analysis and Power BI for visualization, the project transforms transactional data into actionable business insights. The analysis covers overall business performance, outlet and location performance, customer satisfaction, product and category performance, outlet characteristics, and fat-content performance.

The final outcome is an interactive Power BI dashboard supported by SQL-based analysis, business insights, and recommendations. The project provides a comprehensive view of business performance and highlights opportunities for improving inventory planning, product assortment, outlet management, customer experience, and revenue performance.

## 2. Business Requirements

### Objective

The objective of this analysis is to evaluate BlinkIT's sales performance, customer satisfaction, product performance, and outlet performance in order to identify key business insights and opportunities for optimization.

### Project Scope

- Analyze sales performance across different outlet types and locations.
- Evaluate customer satisfaction using average product ratings.
- Identify top-performing product categories and their contribution to total sales.
- Assess how outlet characteristics (size, type, and establishment year) influence sales performance.

| KPI             | Definition                                    |
|-----------------|-----------------------------------------------|
| Total Sales     | Overall revenue generated from all items sold |
| Average Sales   | Average revenue per sale                      |
| Number of Items | Total count of different items sold           |
| Average Rating  | Average customer rating for items sold        |

### Granular Requirements

1. Total Sales by Fat Content
2. Total Sales by Item Type
3. Fat Content by Outlet for Total Sales
4. Total Sales by Outlet Establishment

### 3. Business Insights

#### 3.1 Overall Business KPIs

##### Business Question

What is the overall business performance based on total sales, average sales, number of items, and average customer rating?

##### KPI Results

| KPI             | Result       |
|-----------------|--------------|
| Total Sales     | 1.20 Million |
| Average Sales   | 140.99       |
| Number of Items | 8,523        |
| Average Rating  | 3.97 / 5     |

##### Business Insight

The business generated **1.20 million in total sales** across **8,523 items**, with an average sales value of **140.99 per item**. The overall customer rating is **3.97 out of 5**, indicating a generally positive level of customer satisfaction.

##### Business Interpretation

These results provide a baseline for evaluating the performance of different outlet types, locations, product categories, outlet characteristics, and fat-content segments.

The **3.97 average rating** indicates that customer satisfaction is relatively strong overall, while the remaining gap to a perfect rating suggests potential opportunities for improving product and customer experience.

##### Business Recommendation

- Use **1.20 million in total sales** as the overall sales benchmark.
- Compare individual outlet and product segments against the **140.99 average sales** baseline.
- Investigate the factors influencing the **3.97 average rating** to identify opportunities for improving customer satisfaction.
- Prioritize high-contribution segments while maintaining product quality and customer experience.

## 3.2 Sales Performance by Outlet Type

### Business Question

How does sales performance vary across different outlet types, and how do average sales, item volume, and customer ratings compare?

### KPI Results

| Outlet Type        | Total Sales (M) | Average Sales | Items | Avg. Rating |
|--------------------|-----------------|---------------|-------|-------------|
| Grocery Store      | 0.15            | 140.29        | 1,083 | 3.99        |
| Supermarket Type 1 | 0.79            | 141.21        | 5,577 | 3.96        |
| Supermarket Type 2 | 0.13            | 141.68        | 928   | 3.97        |
| Supermarket Type 3 | 0.13            | 139.80        | 935   | 3.95        |

| OUTLET TYPE       |             |                |           |            |                 |
|-------------------|-------------|----------------|-----------|------------|-----------------|
| Outlet_Type       | Total Sales | Number Of Item | Avg Sales | Avg Rating | Item_Visibility |
| Grocery Store     | 151.94K     | 1083           | 140.29    | 4.0        | 0.10            |
| Supermarket Type2 | 131.48K     | 928            | 141.68    | 4.0        | 0.06            |
| Supermarket Type1 | 787.55K     | 5577           | 141.21    | 4.0        | 0.06            |
| Supermarket Type3 | 130.71K     | 935            | 139.80    | 4.0        | 0.06            |
| Total             | 1,201.68K   | 8523           | 140.99    | 4.0        | 0.07            |

### Business Insight

**Supermarket Type 1** is the strongest outlet type by total sales, generating **0.79 million** and accounting for the highest item volume at **5,577 items**.

Meanwhile, **Supermarket Type 2** records the highest average sales at **141.68**, despite having only **928 items**. Grocery Stores achieve the highest average rating at **3.99**, while Supermarket Type 3 records the lowest at **3.95**.

### Business Interpretation

The strong total sales performance of Supermarket Type 1 appears to be primarily associated with its substantially higher item volume rather than a significantly higher average sales value.

Average sales across outlet types are relatively close, ranging from **139.80 to 141.68**. This indicates that **sales volume is a more prominent differentiator than average sales per item** when comparing outlet types.

### **Business Recommendation**

- Prioritize **Supermarket Type 1** for inventory planning and product availability.
- Investigate the factors enabling **Supermarket Type 2** to achieve the highest average sales.
- Use the **3.99 rating of Grocery Stores** as a customer satisfaction benchmark.
- Investigate the lower **3.95 rating of Supermarket Type 3** for potential customer-experience improvements.

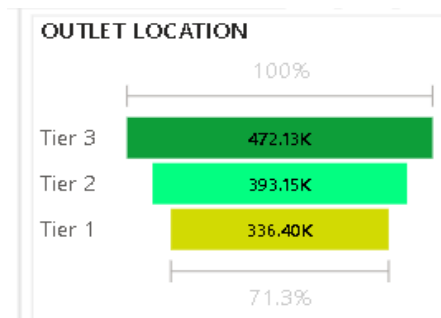
### 3.3 Sales Performance by Outlet Location Type

#### Business Question

How does sales performance vary across different outlet location types?

#### KPI Results

| Outlet Location | Total Sales (M) | Average Sales | Items | Avg. Rating |
|-----------------|-----------------|---------------|-------|-------------|
| Tier 3          | 0.47            | 140.94        | 3,350 | 3.96        |
| Tier 2          | 0.39            | 141.17        | 2,785 | 3.96        |
| Tier 1          | 0.34            | 140.87        | 2,388 | 3.98        |



#### Business Insight

**Tier 3 locations** generate the highest total sales at **0.47 million**, supported by the highest item volume of **3,350 items**. Tier 2 follows with **0.39 million**, while Tier 1 generates **0.34 million**.

Average sales are highly consistent across location tiers, ranging from **140.87 to 141.17**.

Tier 1 has the highest average rating at **3.98**, while Tier 2 and Tier 3 both record **3.96**.

#### Business Interpretation

The higher total sales generated by Tier 3 locations appear to be primarily associated with their higher item volume rather than a materially higher average sales value.

The narrow difference in average sales across tiers indicates that **sales volume is more important than average sales per item** in explaining the difference in total sales.

#### Business Recommendation

- Prioritize **Tier 3 locations** for inventory planning due to their higher sales and item volume.
- Investigate the demand and operational factors contributing to stronger Tier 3 volume.
- Use the **3.98 rating in Tier 1** as a customer-experience benchmark.
- Allocate resources based on actual sales volume rather than assuming higher-tier locations automatically generate higher sales.

### 3.4 Customer Satisfaction by Item Type

#### Business Question

How does customer satisfaction vary across product categories, and how do ratings relate to item volume and sales performance?

#### KPI Results

| Item Type             | Avg. Rating | Items | Total Sales (M) |
|-----------------------|-------------|-------|-----------------|
| Meat                  | 4.02        | 425   | 0.06            |
| Household             | 4.00        | 910   | 0.14            |
| Health and Hygiene    | 3.99        | 520   | 0.07            |
| Canned                | 3.99        | 649   | 0.09            |
| Baking Goods          | 3.98        | 648   | 0.08            |
| Frozen Foods          | 3.97        | 856   | 0.12            |
| Dairy                 | 3.97        | 682   | 0.10            |
| Fruits and Vegetables | 3.96        | 1,232 | 0.18            |
| Seafood               | 3.96        | 64    | 0.01            |
| Snack Foods           | 3.95        | 1,200 | 0.18            |
| Others                | 3.95        | 169   | 0.02            |
| Breakfast             | 3.93        | 110   | 0.02            |
| Soft Drinks           | 3.92        | 445   | 0.06            |
| Starchy Foods         | 3.92        | 148   | 0.02            |
| Hard Drinks           | 3.91        | 214   | 0.03            |
| Breads                | 3.88        | 251   | 0.04            |

#### Business Insight

**Meat** has the highest average customer rating at **4.02**, while **Breads** has the lowest at **3.88**.

However, the highest-rated category is not necessarily the highest-selling category. **Fruits and Vegetables** and **Snack Foods** each generate **0.18 million** in sales, supported by **1,232** and **1,200 items**, respectively.

## Business Interpretation

Customer satisfaction and sales performance do not appear to move proportionally.

For example, Meat has the highest rating at **4.02** but generates only **0.06 million** in sales, while Snack Foods has a lower rating of **3.95** but generates **0.18 million**.

Therefore, **customer satisfaction and commercial performance should be evaluated as separate dimensions** when assessing product performance.

## Business Recommendation

- Maintain the quality of highly rated categories such as **Meat**.
- Prioritize inventory availability for high-volume categories such as **Fruits and Vegetables** and **Snack Foods**.
- Investigate **Breads**, which has the lowest rating at **3.88**, for potential quality, assortment, or customer-experience issues.
- Avoid using customer rating alone to determine inventory or promotional priorities.



### 3.5 Customer Satisfaction by Outlet Type

#### Business Question

How does customer satisfaction vary across outlet types, and how does it compare with sales performance and item volume?

#### KPI Results

| Outlet Type        | Avg. Rating | Items | Total Sales (M) |
|--------------------|-------------|-------|-----------------|
| Grocery Store      | 3.99        | 1,083 | 0.15            |
| Supermarket Type 2 | 3.97        | 928   | 0.13            |
| Supermarket Type 1 | 3.96        | 5,577 | 0.79            |
| Supermarket Type 3 | 3.95        | 935   | 0.13            |

#### Business Insight

Grocery Stores achieve the highest average customer rating at **3.99**, despite generating only **0.15 million** in sales.

Supermarket Type 1 generates the highest total sales at **0.79 million** and has the highest item volume at **5,577**, but its average rating is slightly lower at **3.96**.

Supermarket Type 3 records the lowest average rating at **3.95**.

#### Business Interpretation

The results indicate that higher sales volume does not necessarily correspond to higher customer satisfaction.

Supermarket Type 1 demonstrates strong commercial scale, while Grocery Stores achieve a higher customer rating. This suggests that **sales performance and customer experience should be monitored as complementary but separate performance dimensions**.

#### Business Recommendation

- Use Grocery Stores' **3.99 rating** as a customer satisfaction benchmark.
- Maintain Supermarket Type 1's strong sales performance while identifying opportunities to improve its customer rating.
- Review the customer experience at Supermarket Type 3.
- Identify practices used by higher-rated outlet types that could be applied to lower-rated outlets.

### 3.6 Top-Performing Product Categories

#### Business Question

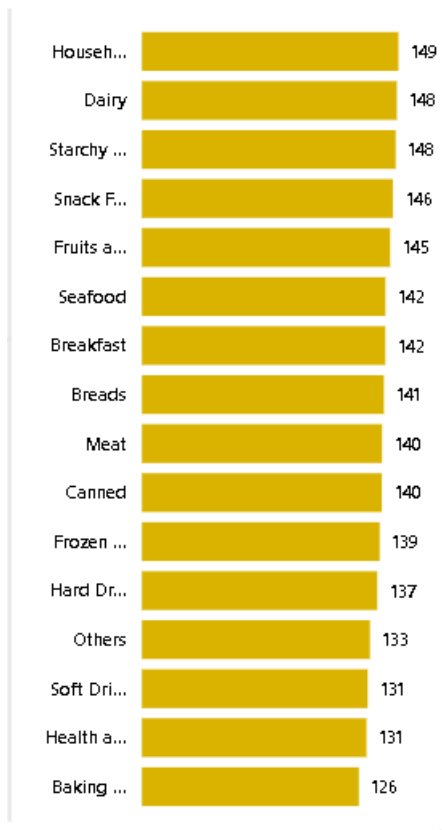
Which product categories are the top performers based on total sales, and how do average sales, item volume, and customer ratings compare?

#### KPI Results

| Item Type             | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|-----------------------|-----------------|------------|-------|-------------|
| Fruits and Vegetables | 0.18            | 144.58     | 1,232 | 3.96        |
| Snack Foods           | 0.18            | 146.19     | 1,200 | 3.95        |
| Household             | 0.14            | 149.42     | 910   | 4.00        |
| Frozen Foods          | 0.12            | 138.50     | 856   | 3.97        |
| Dairy                 | 0.10            | 148.50     | 682   | 3.97        |
| Canned                | 0.09            | 139.76     | 649   | 3.99        |
| Baking Goods          | 0.08            | 126.38     | 648   | 3.98        |
| Health and Hygiene    | 0.07            | 130.82     | 520   | 3.99        |
| Soft Drinks           | 0.06            | 131.49     | 445   | 3.92        |
| Meat                  | 0.06            | 139.88     | 425   | 4.02        |
| Breads                | 0.04            | 140.95     | 251   | 3.88        |
| Hard Drinks           | 0.03            | 137.08     | 214   | 3.91        |
| Others                | 0.02            | 132.85     | 169   | 3.95        |
| Breakfast             | 0.02            | 141.79     | 110   | 3.93        |
| Starchy Foods         | 0.02            | 147.84     | 148   | 3.92        |
| Seafood               | 0.01            | 141.84     | 64    | 3.96        |

## ITEM TYPE

## Business Insight



**Fruits and Vegetables** and **Snack Foods** are the top-performing categories by total sales, each generating **0.18 million**. They also have the highest item volumes, with **1,232** and **1,200 items**, respectively.

However, **Household** has the highest average sales at **149.42** and maintains a strong **4.00 rating**, indicating strong performance despite lower item volume.

## Business Interpretation

The results suggest that higher total sales are strongly associated with higher item volume.

Fruits and Vegetables and Snack Foods lead total sales primarily because of their substantially higher item volumes. Meanwhile, Household demonstrates a different strength through its combination of **high average sales and strong customer satisfaction**.

Therefore, product performance should be evaluated using multiple dimensions rather than total sales alone.

## Business Recommendation

- Prioritize **Fruits and Vegetables** and **Snack Foods** for inventory availability.
- Investigate **Household** as a potentially high-value category because of its **149.42 average sales** and **4.00 rating**.
- Maintain product quality in Meat, which has the highest rating at **4.02**.
- Review Breads, which has the lowest rating at **3.88**.
- Use sales volume, average sales, and customer satisfaction together when making assortment and promotional decisions.

### 3.7 Sales Contribution by Product Category

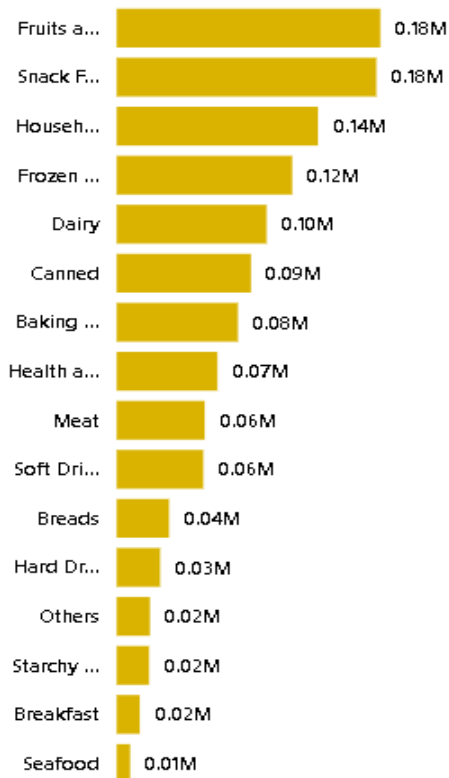
#### Business Question

Which product categories contribute the most to total sales, and how concentrated is revenue across categories?

#### KPI Results

| Item Type             | Total Sales (M) | Sales Contribution |
|-----------------------|-----------------|--------------------|
| Fruits and Vegetables | 0.18            | 14.82%             |
| Snack Foods           | 0.18            | 14.60%             |
| Household             | 0.14            | 11.32%             |
| Frozen Foods          | 0.12            | 9.87%              |
| Dairy                 | 0.10            | 8.43%              |
| Canned                | 0.09            | 7.55%              |
| Baking Goods          | 0.08            | 6.82%              |
| Health and Hygiene    | 0.07            | 5.66%              |
| Meat                  | 0.06            | 4.95%              |
| Soft Drinks           | 0.06            | 4.87%              |
| Breads                | 0.04            | 2.94%              |
| Hard Drinks           | 0.03            | 2.44%              |
| Others                | 0.02            | 1.87%              |
| Starchy Foods         | 0.02            | 1.82%              |
| Breakfast             | 0.02            | 1.30%              |
| Seafood               | 0.01            | 0.76%              |

#### ITEM TYPE



#### Business Insight

**Fruits and Vegetables** contribute the largest share of total sales at **14.82%**, followed by **Snack Foods** at **14.60%**.

Together, these two categories contribute **29.42%** of total sales, meaning nearly one-third of revenue comes from only two categories.

**Household** ranks third with an **11.32%** contribution, while **Seafood** contributes the smallest share at **0.76%**.

#### Business Interpretation

Sales are relatively concentrated among the leading product categories.

The top three categories — **Fruits and Vegetables**, **Snack Foods**, and **Household** — contribute:

**14.82% + 14.60% + 11.32% = 40.74%** of total sales.

This means that the performance of a relatively small number of major categories has a substantial influence on overall business revenue.

#### Business Recommendation

- Prioritize **Fruits and Vegetables** and **Snack Foods**, which together contribute **29.42%** of total sales.
- Ensure strong inventory availability in high-contribution categories to reduce potential lost sales.
- Maintain **Household** as an important revenue contributor.
- Review low-contribution categories such as **Seafood** to determine whether assortment optimization or targeted strategies are appropriate.
- Avoid eliminating low-contribution categories based solely on sales contribution; consider customer demand and strategic importance as well.

## 3.8 Sales Performance by Outlet Size

### Business Question

How does outlet size influence sales performance, item volume, and customer satisfaction?

### KPI Results

| Outlet Size | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|-------------|-----------------|------------|-------|-------------|
| Medium      | 0.51            | 139.88     | 3,631 | 3.98        |
| Small       | 0.44            | 141.70     | 3,139 | 3.96        |
| High        | 0.25            | 142.04     | 1,753 | 3.95        |

### Business Insight

Medium-sized outlets generate the highest total sales at **0.51 million**, supported by the highest item volume of **3,631 items**.

High-sized outlets have the highest average sales at **142.04**, but generate the lowest total sales at **0.25 million** because they have the lowest item volume.

Medium-sized outlets also achieve the highest customer rating at **3.98**, while High-sized outlets record the lowest at **3.95**.

### Business Interpretation

Outlet size alone does not guarantee higher total sales.

Medium-sized outlets outperform High-sized outlets in total sales because of their substantially higher item volume. High-sized outlets achieve a slightly higher average sales value but significantly lower volume.

Therefore, sales volume appears to be a stronger factor than average sales per item when comparing total outlet performance.

### Business Recommendation

- Prioritize Medium-sized outlets for inventory availability due to their highest sales and item volume.
- Investigate why High-sized outlets have lower item volume despite having the highest average sales.
- Use the **3.98 rating of Medium outlets** as a customer satisfaction benchmark.
- Evaluate whether High-sized outlets require improvements in product assortment, customer traffic, or inventory utilization.

### 3.9 Outlet Type Performance Comparison

#### Business Question

How do outlet types differ in overall sales performance, sales efficiency, item volume, and customer satisfaction?

#### KPI Results

| Outlet Type        | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|--------------------|-----------------|------------|-------|-------------|
| Supermarket Type 1 | 0.79            | 141.21     | 5,577 | 3.96        |
| Grocery Store      | 0.15            | 140.29     | 1,083 | 3.99        |
| Supermarket Type 2 | 0.13            | 141.68     | 928   | 3.97        |
| Supermarket Type 3 | 0.13            | 139.80     | 935   | 3.95        |

| OUTLET TYPE       |             |                |           |            |                 |
|-------------------|-------------|----------------|-----------|------------|-----------------|
| Outlet_Type       | Total Sales | Number Of Item | Avg Sales | Avg Rating | Item_Visibility |
| Grocery Store     | 151.94K     | 1083           | 140.29    | 4.0        | 0.10            |
| Supermarket Type2 | 131.48K     | 928            | 141.68    | 4.0        | 0.06            |
| Supermarket Type1 | 787.55K     | 5577           | 141.21    | 4.0        | 0.06            |
| Supermarket Type3 | 130.71K     | 935            | 139.80    | 4.0        | 0.06            |
| Total             | 1,201.68K   | 8523           | 140.99    | 4.0        | 0.07            |

#### Business Insight

Supermarket Type 1 is the strongest outlet type by total sales, generating **0.79 million** with **5,577 items**.

Supermarket Type 2 achieves the highest average sales at **141.68**, while Grocery Stores record the highest customer rating at **3.99**.

Supermarket Type 3 has both the lowest average sales at **139.80** and the lowest customer rating at **3.95**.

## Business Interpretation

Supermarket Type 1's strong total sales performance is primarily associated with its substantially higher item volume rather than the highest average sales per item.

The results also highlight a distinction between **scale and customer satisfaction**: Grocery Stores have the highest rating but substantially lower sales volume, while Supermarket Type 1 generates significantly higher sales with a slightly lower rating.

## Business Recommendation

- Continue prioritizing Supermarket Type 1 as the primary sales contributor.
- Investigate the operational factors behind its high item volume.
- Examine Supermarket Type 2 to understand how it achieves the highest average sales.
- Use Grocery Stores' **3.99 rating** as a customer satisfaction benchmark.
- Investigate Supermarket Type 3's lower rating and average sales for potential operational improvements.



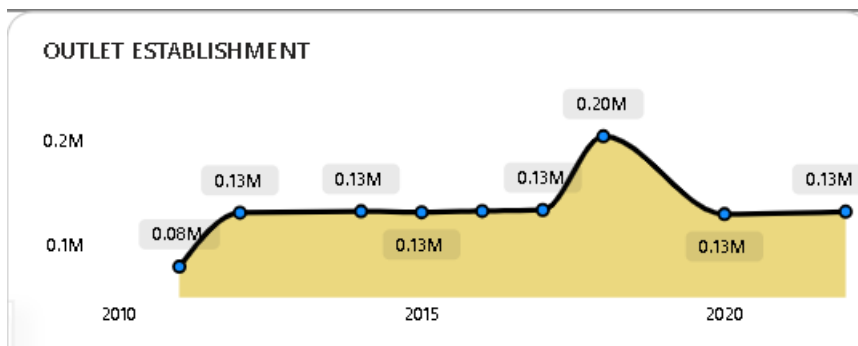
### 3.10 Sales Performance by Outlet Establishment Year

#### Business Question

How does outlet establishment year influence sales performance, item volume, and customer satisfaction?

#### KPI Results

| Establishment Year | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|--------------------|-----------------|------------|-------|-------------|
| 2011               | 0.08            | 140.78     | 555   | 3.98        |
| 2012               | 0.13            | 140.30     | 930   | 3.99        |
| 2014               | 0.13            | 141.43     | 932   | 3.95        |
| 2015               | 0.13            | 140.95     | 929   | 3.96        |
| 2016               | 0.13            | 142.06     | 930   | 3.96        |
| 2017               | 0.13            | 143.12     | 930   | 3.94        |
| 2018               | 0.20            | 139.80     | 1,463 | 3.97        |
| 2020               | 0.13            | 139.42     | 926   | 3.98        |
| 2022               | 0.13            | 141.68     | 928   | 3.97        |



#### Business Insight

Outlets established in **2018** generate the highest total sales at **0.20 million**, together with the highest item volume of **1,463 items**.

The **2017** group has the highest average sales at **143.12**, despite generating only **0.13 million** in total sales.

Meanwhile, outlets established in **2012** achieve the highest average rating at **3.99**, while the 2017 group records the lowest at **3.94**.

## Business Interpretation

The results do not indicate a consistent relationship between establishment year and total sales performance.

The 2018 group stands out because its higher total sales are accompanied by substantially higher item volume. In contrast, the 2017 group achieves the highest average sales but not the highest total sales because its item volume is much lower than that of the 2018 group.

This suggests that **sales volume and outlet scale appear more relevant to total sales than establishment year alone.**

## Business Recommendation

- Investigate the characteristics of 2018 outlets that contribute to their higher sales and item volume.
- Analyze the practices of 2017 outlets that support the highest average sales.
- Use the **3.99 rating of 2012 outlets** as a customer satisfaction benchmark.
- Investigate the lower **3.94 rating of 2017 outlets**.
- Avoid making expansion or investment decisions based solely on establishment year; combine it with outlet size, type, location, and sales volume.

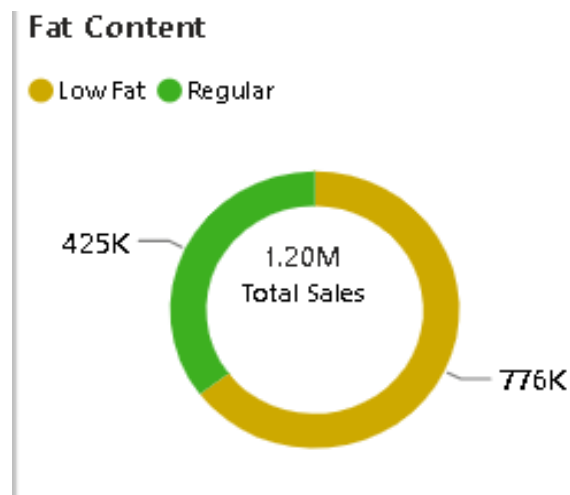
### 3.11 Sales Performance by Fat Content

#### Business Question

How does sales performance differ between Low Fat and Regular products, and how do average sales, item volume, and customer ratings compare?

#### KPI Results

| Fat Content | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|-------------|-----------------|------------|-------|-------------|
| Low Fat     | 0.78            | 140.71     | 5,517 | 3.97        |
| Regular     | 0.43            | 141.50     | 3,006 | 3.97        |



#### Business Insight

Low Fat products generate substantially higher total sales at **0.78 million**, compared with **0.43 million** for Regular products.

Low Fat products also have significantly higher item volume, with **5,517 items** versus **3,006**.

Regular products have slightly higher average sales at **141.50**, compared with **140.71** for Low Fat products. Both categories have the same average rating of **3.97**.

#### Business Interpretation

The stronger total sales performance of Low Fat products appears to be driven primarily by **higher item volume rather than higher average sales per item**.

The identical customer ratings indicate no meaningful difference in overall customer satisfaction between the two fat-content categories based on this dataset.

## Business Recommendation

- Prioritize Low Fat products in inventory planning due to their higher sales and item volume.
- Maintain sufficient availability of Regular products, which still generate **0.43 million** in sales.
- Focus product strategy primarily on demand and volume rather than rating differences.
- Investigate whether the stronger Low Fat demand is consistent across different outlet types and locations.

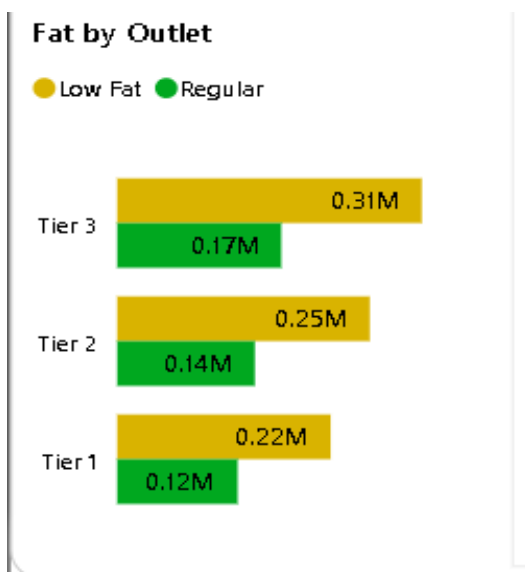
### 3.12 Fat Content Performance by Outlet Type

#### Business Question

How do Low Fat and Regular product sales vary across outlet types, and what differences exist in average sales, item volume, and customer ratings?

#### KPI Results

| Outlet Type        | Fat Content | Total Sales (M) | Avg. Sales | Items | Avg. Rating |
|--------------------|-------------|-----------------|------------|-------|-------------|
| Grocery Store      | Low Fat     | 0.10            | 141.58     | 705   | 4.00        |
| Grocery Store      | Regular     | 0.05            | 137.90     | 378   | 3.97        |
| Supermarket Type 1 | Low Fat     | 0.51            | 140.73     | 3,609 | 3.96        |
| Supermarket Type 1 | Regular     | 0.28            | 142.11     | 1,968 | 3.96        |
| Supermarket Type 2 | Low Fat     | 0.08            | 141.88     | 598   | 3.97        |
| Supermarket Type 2 | Regular     | 0.05            | 141.31     | 330   | 3.98        |
| Supermarket Type 3 | Low Fat     | 0.08            | 138.47     | 605   | 3.95        |
| Supermarket Type 3 | Regular     | 0.05            | 142.24     | 330   | 3.96        |



## Business Insight

Low Fat products generate higher total sales than Regular products across **every outlet type**.

The largest difference occurs in Supermarket Type 1, where Low Fat products generate **0.51 million** compared with **0.28 million** for Regular products. Supermarket Type 1 also records the highest Low Fat item volume at **3,609 items**.

However, Regular products achieve higher average sales in Supermarket Type 1, Supermarket Type 3, and Grocery Stores. The largest difference occurs in Supermarket Type 3, where Regular products achieve **142.24** compared with **138.47** for Low Fat products.

## Business Interpretation

The stronger total sales of Low Fat products are primarily driven by **higher item volume rather than higher average sales per item**.

Low Fat products perform particularly strongly in Supermarket Type 1, which is also the highest-volume outlet type in the analysis.

Customer ratings remain relatively close across all combinations. The highest rating is **4.00** for Low Fat products in Grocery Stores, while the lowest is **3.95** for Low Fat products in Supermarket Type 3.

## Business Recommendation

- Prioritize Low Fat products in inventory planning, particularly in **Supermarket Type 1**, where they generate **0.51 million** in sales.
- Maintain Regular product availability because Regular products achieve higher average sales in several outlet types.
- Use the **4.00 rating of Low Fat products in Grocery Stores** as a customer satisfaction benchmark.
- Investigate the **3.95 rating of Low Fat products in Supermarket Type 3**.
- Consider outlet-specific product assortment strategies rather than applying the same Low Fat/Regular mix across all outlet types.

## 4. Key Business Takeaways

The analysis highlights several major patterns across BlinkIT's business performance.

- **Sales Volume Is a Major Driver of Revenue**

The strongest total sales are generally associated with higher item volume rather than significantly higher average sales per item.

This pattern is particularly visible across **Supermarket Type 1, Tier 3 locations, Medium-sized outlets, and the 2018 establishment group**.

- **Supermarket Type 1 Is the Primary Sales Contributor**

Supermarket Type 1 generates **0.79 million** in sales and accounts for **5,577 items**, making it the strongest outlet type by total sales and item volume.

- **Product Revenue Is Concentrated**

**Fruits and Vegetables, Snack Foods, and Household** together contribute **40.74% of total sales**, indicating that several major product categories have a significant influence on overall revenue.

- **Customer Satisfaction and Sales Are Not Directly Proportional**

The highest-rated categories and outlets are not necessarily the highest-selling ones.

For example, Meat achieves the highest product rating at **4.02**, while Fruits and Vegetables and Snack Foods generate the highest category sales.

- **Low Fat Products Have Stronger Demand**

Low Fat products generate **0.78 million** in sales compared with **0.43 million** for Regular products.

The difference is primarily associated with higher item volume rather than higher average sales per item.

- **Outlet Size Does Not Automatically Determine Sales**

Medium-sized outlets generate the highest total sales at **0.51 million**, while High-sized outlets have the highest average sales but the lowest total sales.

- **Operational Strategy Should Be Segment-Specific**

The results indicate that inventory, assortment, customer experience, and promotional strategies should be adapted according to **outlet type, location, product category, outlet characteristics, and customer demand** rather than applying a single strategy across all segments.



## 5. Conclusion

The BlinkIT Grocery Sales Analysis provides a comprehensive view of business performance using **SQL (MySQL)** and **Power BI**.

The analysis demonstrates that **sales volume, product category, outlet characteristics, and customer satisfaction should be evaluated together** to understand business performance effectively.

Supermarket Type 1, Fruits and Vegetables, Snack Foods, Medium-sized outlets, and Low Fat products emerge as important contributors to overall sales. At the same time, customer satisfaction analysis identifies opportunities to improve lower-rated product categories and outlet types.

Overall, the findings provide a data-driven foundation for **inventory planning, product assortment, outlet management, customer experience improvement, and revenue optimization**.

The project demonstrates an end-to-end analytical workflow, from SQL-based business analysis to Power BI visualization and business-oriented recommendations.